

Linear regression

6 Exercises

The exercise numbering is the same as in the lecture notes. References to earlier sections and equations refer to those notes.

The first three exercises and Exercise 6 use the dataset below. The exercises can be done by hand with a calculator. Use the sum of squared errors, without a factor of $1/2$ or $1/N$, unless another objective is explicitly stated.

i	1	2	3	4	5
x_i	-2	-1	0	1	2
y_i	-6.2	-2.6	0.5	2.7	5.7

1. Compare three predictors.

Compute the predictions and sum-of-squared-errors value for each function

$$f_1(x) = x, \quad f_2(x) = 2x, \quad f_3(x) = 3x.$$

Which gives the smallest loss on the displayed dataset? Does this establish that it is the best predictor among all straight lines?

2. Fit a straight line.

Collect the intercept and slope in the column vector $\boldsymbol{\theta} = (b, a)^\top$, and fit the family

$$f_{\boldsymbol{\theta}}^{\text{aff}}(x) = b + ax$$

by squared-error minimisation, using the matrix method in Section 4. Construct $\mathbf{X}$ with a first column of ones and a second column containing the inputs. Then compute $\mathbf{X}^\top \mathbf{X}$ and $\mathbf{X}^\top \mathbf{y}$, and solve the normal equations for $\boldsymbol{\theta}$. Check that the minimiser is unique.

3. Fit a quadratic without an intercept.

Collect the two coefficients in the column vector $\boldsymbol{\theta} = (w_1, w_2)^\top$, and fit the predictor

$$f_{\boldsymbol{\theta}}(x) := w_1 x + w_2 x^2$$

by the same squared-error criterion. Specify the two feature functions, form $\boldsymbol{\Phi}$, and compute the fitted coefficients. This family has no constant term, so do not add a column of ones. Check that $\boldsymbol{\Phi}^\top \boldsymbol{\Phi}$ is invertible, as required to apply formula (52).

4. Differentiate the absolute value.

Let $q : \mathbb{R} \rightarrow \mathbb{R}$ be $q(s) = |s|$. Sketch its graph and find $q'(s)$ separately for $s < 0$ and $s > 0$. Explain why the derivative does not exist at zero. Express the derivative at nonzero inputs using

$$\text{sign}(s) = \begin{cases} -1, & s < 0, \\ 0, & s = 0, \\ 1, & s > 0. \end{cases} \quad (74)$$

The convention $\text{sign}(0) = 0$ in (74) is not a statement that $q'(0)$ exists.

5. Changing the loss for the constant predictor.

Return to the five observations used in the worked examples:

i	1	2	3	4	5
x_i	-2	-1	0	1	2
y_i	-1.5	0.5	1	2.5	7.5

In Section 3.1, we fitted the constant predictor from (7) using the squared loss ℓ_{sq} from (4). We found that the best constant is the mean $\bar{y} = 2$, with $\mathcal{L}_{\text{sq}}^{\text{const}}(2) = 46$.

What changes if we use the absolute loss ℓ_{abs} from (4) instead? Keeping the same data and family of predictors, we now minimise

$$\mathcal{L}_{\text{abs}}^{\text{const}}(b) = \sum_{i=1}^5 \ell_{\text{abs}}(f_b^{\text{const}}(x_i), y_i) = \sum_{i=1}^5 |b - y_i|. \quad (75)$$

- For b different from all five targets, express the derivative of (75) using the sign function (74) from Exercise 4. Interpret it as the number of targets below b minus the number above it. Find the slope on each interval between successive targets, and on the two outer intervals. Sketch $\mathcal{L}_{\text{abs}}^{\text{const}}$, find its minimiser b_{abs}^* , and relate it to the median of the five targets. Does $(\mathcal{L}_{\text{abs}}^{\text{const}})'(b_{\text{abs}}^*)$ exist?
- Compare your fitted constant with the squared-loss fit already obtained in Section 3.1. Complete the table, where the known squared-error value is supplied:

constant	$\mathcal{L}_{\text{abs}}^{\text{const}}(b)$	$\mathcal{L}_{\text{sq}}^{\text{const}}(b)$
b_{abs}^*		
$\bar{y} = 2$		46

Which constant is preferred under each loss? Compare entries within each column, because the two columns measure error differently.

- Plot the five observations and the two horizontal prediction lines $y = b_{\text{abs}}^*$ and $y = \bar{y}$ on the same axes. Explain how changing the loss has changed the optimal predictor, referring to the way the two losses penalise large errors in Section 2.

Reflect on what this comparison shows: choosing a loss determines what we optimise for and can change which predictor is best. If the loss does not reflect the goal of the application, even a predictor that minimises it may be a poor choice for that goal.

Hint for (a). As b crosses one target from left to right, that term's slope changes from -1 to $+1$. At a corner, inspect the slopes on either side, rather than assuming that a minimiser must have a derivative.

6. One gradient-descent update.

Return to the dataset at the beginning of this section and the affine predictor from Exercise 2. Start from

$$\boldsymbol{\theta}^{(0)} = \begin{pmatrix} b^{(0)} \\ a^{(0)} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and take the learning rate $\eta = 0.04$.

Compute the initial residuals and use (64) and (65) to obtain $\boldsymbol{\theta}^{(1)}$. Both coefficient updates must use the same initial residuals. Check your result using the matrix update (70).

Compute the empirical squared loss before and after the update. Has it decreased? Compare the updated coefficients and loss with the minimiser found in Exercise 2.

Solutions

1. Comparing predictions.

The predictions are

x_i	y_i	$f_1(x_i)$	$f_2(x_i)$	$f_3(x_i)$
-2	-6.2	-2	-4	-6
-1	-2.6	-1	-2	-3
0	0.5	0	0	0
1	2.7	1	2	3
2	5.7	2	4	6

Subtract the target from each prediction and square. For each of the three fixed candidates, define the numerical value

$$L_j := \sum_{i=1}^5 (f_j(x_i) - y_i)^2, \quad j \in \{1, 2, 3\}.$$

Evaluating these sums gives

$$\begin{aligned} L_1 &= (4.2)^2 + (1.6)^2 + (-0.5)^2 + (-1.7)^2 + (-3.7)^2 = 37.03, \\ L_2 &= (2.2)^2 + (0.6)^2 + (-0.5)^2 + (-0.7)^2 + (-1.7)^2 = 8.83, \\ L_3 &= (0.2)^2 + (-0.4)^2 + (-0.5)^2 + (0.3)^2 + (0.3)^2 = 0.63. \end{aligned}$$

Thus $f_3(x) = 3x$ is best among the three candidates. We have only compared these three functions, not minimised over all possible slopes and intercepts. Exercise 2 performs that larger optimisation.

2. Fitting the affine model.

The design matrix, coefficient column, and target column are

$$\mathbf{X} = \begin{pmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \boldsymbol{\theta} = \begin{pmatrix} b \\ a \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} -6.2 \\ -2.6 \\ 0.5 \\ 2.7 \\ 5.7 \end{pmatrix}.$$

The sums needed for the matrix products are

$$\sum_{i=1}^5 x_i = 0, \quad \sum_{i=1}^5 x_i^2 = 10, \quad \sum_{i=1}^5 y_i = 0.1, \quad \sum_{i=1}^5 x_i y_i = 29.1.$$

For example, the last sum is $12.4 + 2.6 + 0 + 2.7 + 11.4 = 29.1$. Therefore

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 0 \\ 0 & 10 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{y} = \begin{pmatrix} 0.1 \\ 29.1 \end{pmatrix}.$$

The determinant of the 2×2 matrix is $50 \neq 0$, and

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \begin{pmatrix} 1/5 & 0 \\ 0 & 1/10 \end{pmatrix}.$$

Thus

$$\boldsymbol{\theta}^* = \begin{pmatrix} 1/5 & 0 \\ 0 & 1/10 \end{pmatrix} \begin{pmatrix} 0.1 \\ 29.1 \end{pmatrix} = \begin{pmatrix} 0.02 \\ 2.91 \end{pmatrix}.$$

The unique fitted line is $f_{\theta^*}^{\text{aff}}(x) = 0.02 + 2.91x$. The inverse is of $\mathbf{X}^\top \mathbf{X}$, not of $\mathbf{X} \mathbf{X}^\top$. As a direct check, the residuals are

$$0.40, -0.29, -0.48, 0.23, 0.14.$$

Their sum and their input-weighted sum both equal zero, as the normal equations require. Their squared sum is 0.547, smaller than 0.63. We can also verify the global minimum directly by completing the square:

$$\mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}) = 0.547 + 5(b - 0.02)^2 + 10(a - 2.91)^2.$$

Both squared terms are nonnegative and vanish together only at $(b, a) = (0.02, 2.91)$, so the minimiser is unique.

3. A quadratic without an intercept.

Choose $\phi_1(x) = x$ and $\phi_2(x) = x^2$. Then

$$\boldsymbol{\Phi} = \begin{pmatrix} -2 & 4 \\ -1 & 1 \\ 0 & 0 \\ 1 & 1 \\ 2 & 4 \end{pmatrix}.$$

The products are

$$\boldsymbol{\Phi}^\top \boldsymbol{\Phi} = \begin{pmatrix} \sum_i x_i^2 & \sum_i x_i^3 \\ \sum_i x_i^3 & \sum_i x_i^4 \end{pmatrix} = \begin{pmatrix} 10 & 0 \\ 0 & 34 \end{pmatrix}, \quad \boldsymbol{\Phi}^\top \mathbf{y} = \begin{pmatrix} 29.1 \\ -1.9 \end{pmatrix},$$

where all sums in this solution run from $i = 1$ to 5. In particular,

$$\sum_{i=1}^5 x_i^2 y_i = 4(-6.2) + 1(-2.6) + 0(0.5) + 1(2.7) + 4(5.7) = -1.9.$$

The determinant is $340 \neq 0$, so the least-squares formula $\boldsymbol{\theta}^* = (\boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^\top \mathbf{y}$ applies and the minimiser is unique:

$$\boldsymbol{\theta}^* = \begin{pmatrix} 1/10 & 0 \\ 0 & 1/34 \end{pmatrix} \begin{pmatrix} 29.1 \\ -1.9 \end{pmatrix} = \begin{pmatrix} 2.91 \\ -19/340 \end{pmatrix}.$$

The fitted predictor is therefore

$$f_{\boldsymbol{\theta}^*}(x) = 2.91x - \frac{19}{340}x^2 \approx 2.91x - 0.05588235x^2.$$

The model predicts zero at $x = 0$, because both selected features vanish there. This restriction would change if we added a constant feature, but that is not the family in this exercise.

4. The corner in the absolute value.

For $s < 0$, $q(s) = -s$ and $q'(s) = -1$. For $s > 0$, $q(s) = s$ and $q'(s) = 1$. Thus

$$q'(s) = \text{sign}(s) \quad (s \neq 0).$$

The graph is V-shaped, meeting at the origin.

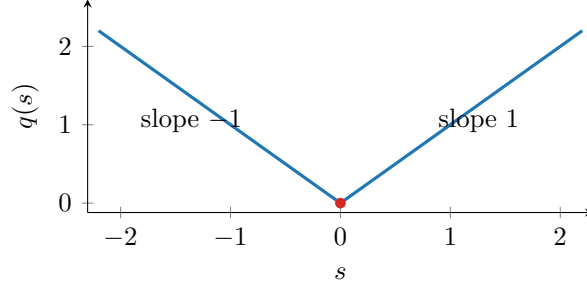


Figure 1: The graph of $q(s) = |s|$ has left-hand slope -1 and right-hand slope 1 at zero, so $q'(0)$ does not exist.

At zero, the difference quotient is

$$\frac{q(h) - q(0)}{h} = \frac{|h|}{h} = \begin{cases} 1, & h > 0, \\ -1, & h < 0. \end{cases}$$

The right and left limits disagree, so no derivative exists there. Defining the separate function sign to have value zero at zero does not change these limits.

5. The same data, two losses.

For squared loss, the fitted constant is the mean of the targets:

$$\bar{y} = \frac{-1.5 + 0.5 + 1 + 2.5 + 7.5}{5} = 2, \quad \mathcal{L}_{\text{sq}}^{\text{const}}(2) = 46.$$

(a) Absolute loss selects the median. For $b \neq y_i$ for all i , differentiating (75) using Exercise 4 gives

$$(\mathcal{L}_{\text{abs}}^{\text{const}})'(b) = \sum_{i=1}^5 \text{sign}(b - y_i).$$

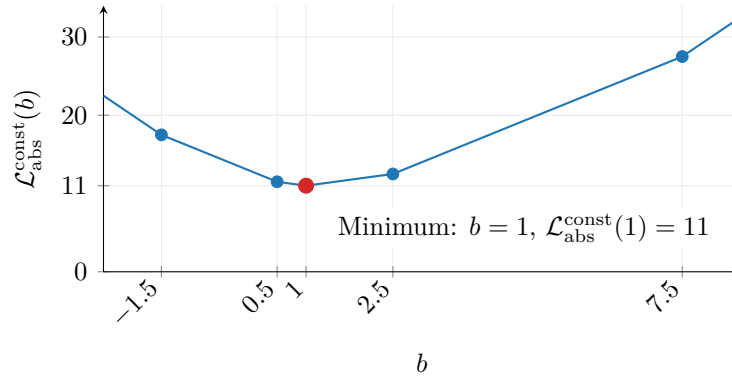
Each target below b contributes $+1$, and each target above it contributes -1 . The targets are already in increasing order, giving

b	$(-\infty, -1.5)$	$(-1.5, 0.5)$	$(0.5, 1)$	$(1, 2.5)$	$(2.5, 7.5)$	$(7.5, \infty)$
$(\mathcal{L}_{\text{abs}}^{\text{const}})'(b)$	-5	-3	-1	1	3	5

Thus $\mathcal{L}_{\text{abs}}^{\text{const}}$ decreases up to $b = 1$ and increases after it. Its unique global minimiser is $b_{\text{abs}}^* = 1$, the middle target: the *median*. The minimum is

$$\mathcal{L}_{\text{abs}}^{\text{const}}(1) = 2.5 + 0.5 + 0 + 1.5 + 6.5 = 11.$$

The slope changes from -1 to $+1$ at this minimum, so $(\mathcal{L}_{\text{abs}}^{\text{const}})'(1)$ does not exist. The minimum is a corner of the graph.



(b) Compare each constant under each loss. Part (a) gives $\mathcal{L}_{\text{abs}}^{\text{const}}(1) = 11$. The two remaining entries are

$$\mathcal{L}_{\text{abs}}^{\text{const}}(2) = 3.5 + 1.5 + 1 + 0.5 + 5.5 = 12,$$

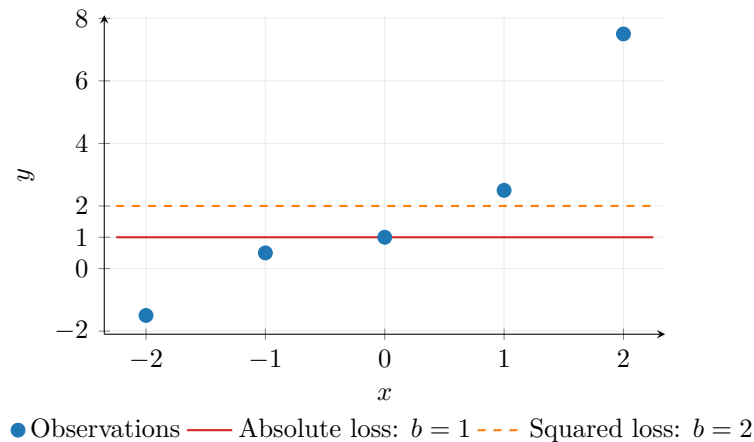
$$\mathcal{L}_{\text{sq}}^{\text{const}}(1) = 2.5^2 + 0.5^2 + 0^2 + (-1.5)^2 + (-6.5)^2 = 51.$$

Hence

constant	$\mathcal{L}_{\text{abs}}^{\text{const}}(b)$	$\mathcal{L}_{\text{sq}}^{\text{const}}(b)$
$b_{\text{abs}}^* = 1$ (median)	11	51
$\bar{y} = 2$ (mean)	12	46

Absolute error prefers $b = 1$, since $11 < 12$. Squared error prefers $b = 2$, since $46 < 51$. Each bold entry is the minimum in its own column, whereas comparing 11 with 46 would compare two different measures of error.

(c) Same data and family, different objectives. The fitted predictors are the horizontal lines $y = 1$ and $y = 2$. The inputs locate the observations in the plot, but do not enter either objective because a constant predictor returns b at every input.



The different fitted constants result from changing how errors are penalised. Doubling a residual's magnitude doubles its absolute penalty but quadruples its squared penalty. Large residuals therefore have more influence under squared error. On these data, moving the constant from 1 to 2 reduces the error at the largest target, 7.5, from 6.5 to 5.5. Its squared penalty falls by 12, outweighing the net increase of 7 in the other squared penalties. The total squared error falls by 5, while the total absolute error rises by 1. The same family thus selects the mean under squared error and the median under absolute error.

6. One gradient-descent update.

At $\boldsymbol{\theta}^{(0)} = (0, 0)^\top$, the predictor is

$$f_{\boldsymbol{\theta}^{(0)}}^{\text{aff}}(x) = b^{(0)} + a^{(0)}x = 0.$$

Therefore, each initial residual is the negative of its target:

$$r_i^{(0)} = b^{(0)} + a^{(0)}x_i - y_i = -y_i, \quad i = 1, \dots, 5.$$

For the dataset in this exercise, this gives

$$\mathbf{r}^{(0)} = \begin{pmatrix} 6.2 \\ 2.6 \\ -0.5 \\ -2.7 \\ -5.7 \end{pmatrix}.$$

The sums needed for the two coefficient updates are

$$\sum_{i=1}^5 r_i^{(0)} = -\sum_{i=1}^5 y_i = -0.1, \quad \sum_{i=1}^5 x_i r_i^{(0)} = -\sum_{i=1}^5 x_i y_i = -29.1.$$

Using these sums in the two coefficient updates, with $\eta = 0.04$, gives

$$\begin{aligned} b^{(1)} &= b^{(0)} - 2\eta \sum_{i=1}^5 r_i^{(0)} = 0 - 2(0.04)(-0.1) = 0.008, \\ a^{(1)} &= a^{(0)} - 2\eta \sum_{i=1}^5 x_i r_i^{(0)} = 0 - 2(0.04)(-29.1) = 2.328. \end{aligned}$$

Both values are computed from the residuals of the initial line. We do not recompute the residuals between the two coefficient updates.

To check the calculation in matrix form, use the design matrix from Exercise 2. Then

$$\mathbf{X}^\top \mathbf{r}^{(0)} = \begin{pmatrix} \sum_{i=1}^5 r_i^{(0)} \\ \sum_{i=1}^5 x_i r_i^{(0)} \end{pmatrix} = \begin{pmatrix} -0.1 \\ -29.1 \end{pmatrix}.$$

Consequently, the matrix form of the parameter update gives

$$\boldsymbol{\theta}^{(1)} = \boldsymbol{\theta}^{(0)} - 2\eta \mathbf{X}^\top \mathbf{r}^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 2(0.04) \begin{pmatrix} -0.1 \\ -29.1 \end{pmatrix} = \begin{pmatrix} 0.008 \\ 2.328 \end{pmatrix},$$

in agreement with the coordinate calculations.

The updated predictor is therefore

$$f_{\boldsymbol{\theta}^{(1)}}^{\text{aff}}(x) = 0.008 + 2.328x.$$

We now recompute its predictions and residuals:

x_i	y_i	$f_{\boldsymbol{\theta}^{(1)}}^{\text{aff}}(x_i)$	$r_i^{(1)}$
-2	-6.2	-4.648	1.552
-1	-2.6	-2.320	0.280
0	0.5	0.008	-0.492
1	2.7	2.336	-0.364
2	5.7	4.664	-1.036

Before the update, the empirical squared loss was

$$\mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^{(0)}) = 6.2^2 + 2.6^2 + (-0.5)^2 + (-2.7)^2 + (-5.7)^2 = 85.23.$$

After the update, it is

$$\begin{aligned}\mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^{(1)}) &= 1.552^2 + 0.280^2 + (-0.492)^2 + (-0.364)^2 + (-1.036)^2 \\ &= 3.93496.\end{aligned}$$

Thus the loss has decreased. However, Exercise 2 gave the unique minimiser

$$\boldsymbol{\theta}^* = \begin{pmatrix} 0.02 \\ 2.91 \end{pmatrix}, \quad \mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^*) = 0.547.$$

Therefore, this first update has improved the fit, but it has not reached the minimiser. To perform another iteration, we would use the new residuals $\mathbf{r}^{(1)}$ to update both coefficients again.